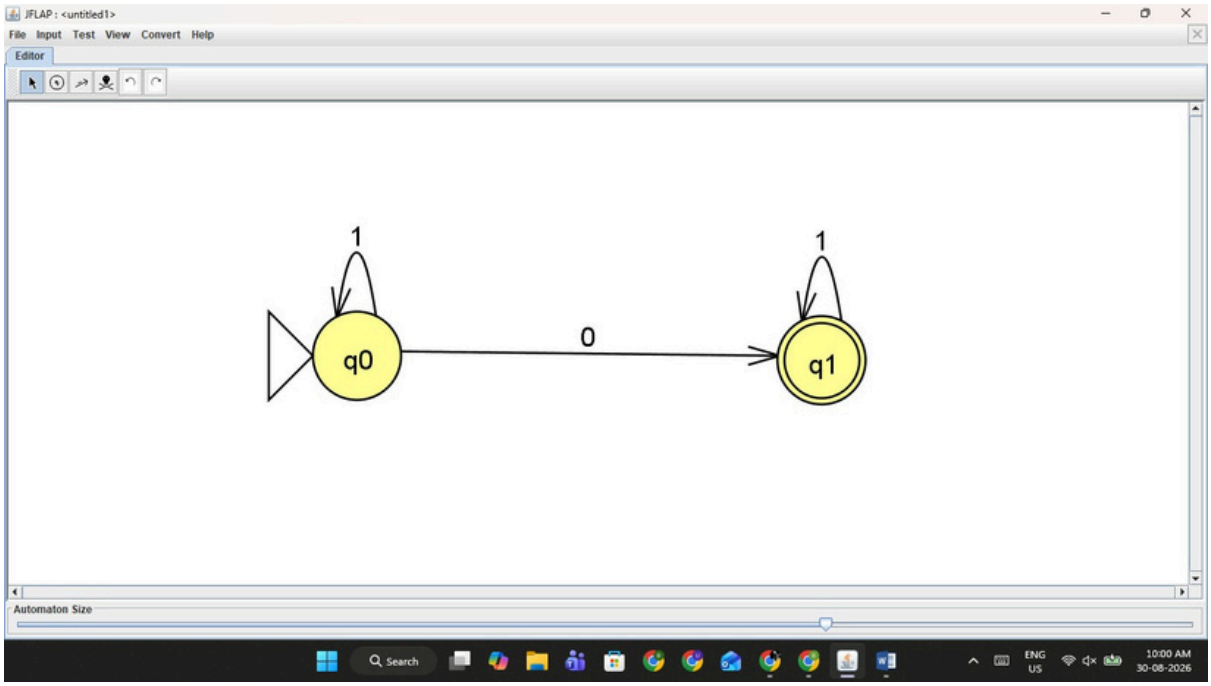
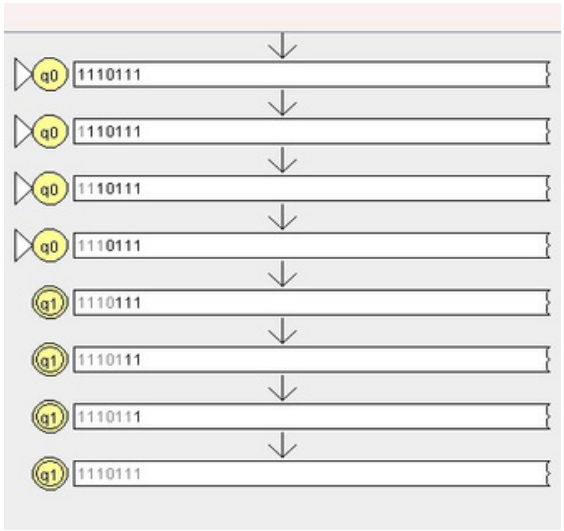
JFLAP Experiment

1. Design a DFA that accepts strings which contain 0 in it over the alphabet Σ = { 0 , 1 }

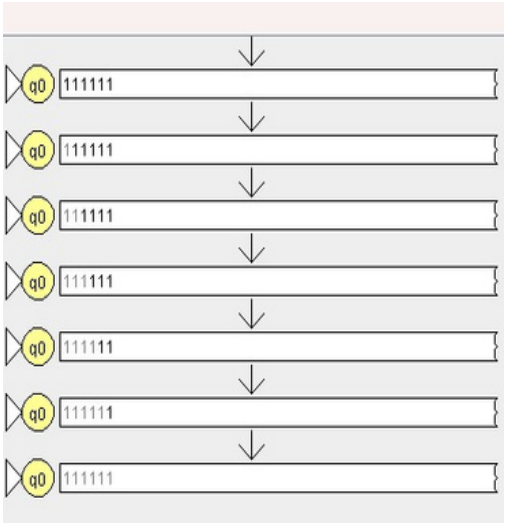


Input: 1110111



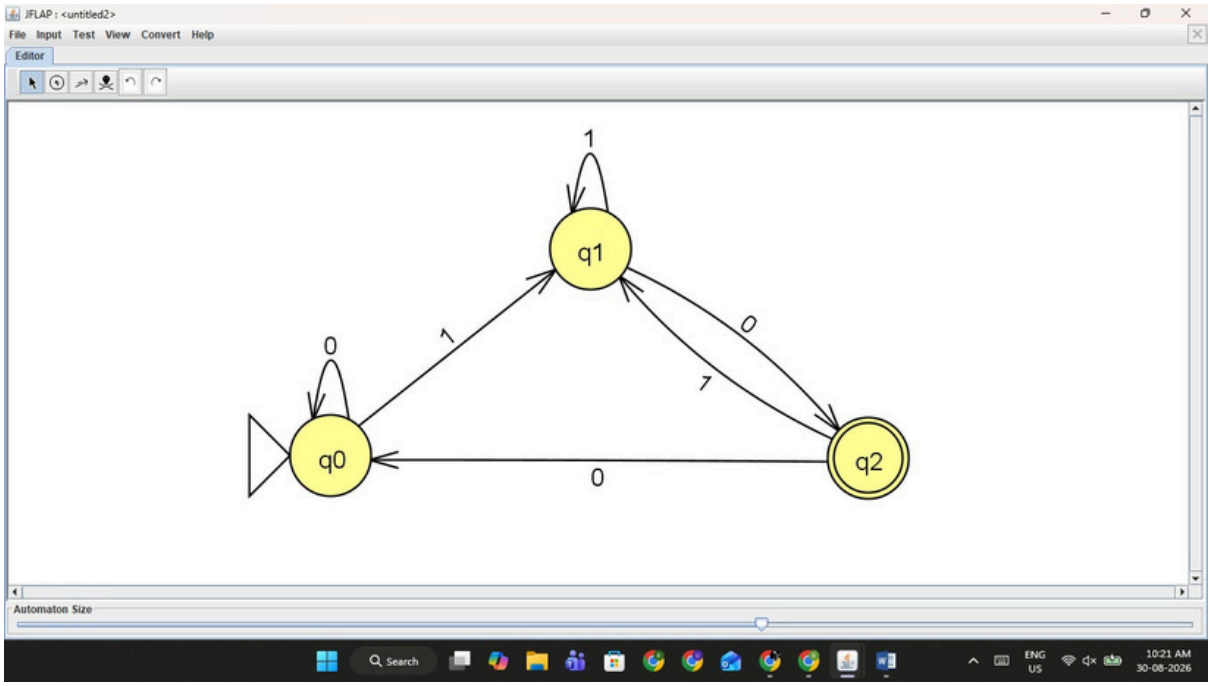
Result: Accepted

Input: 111111

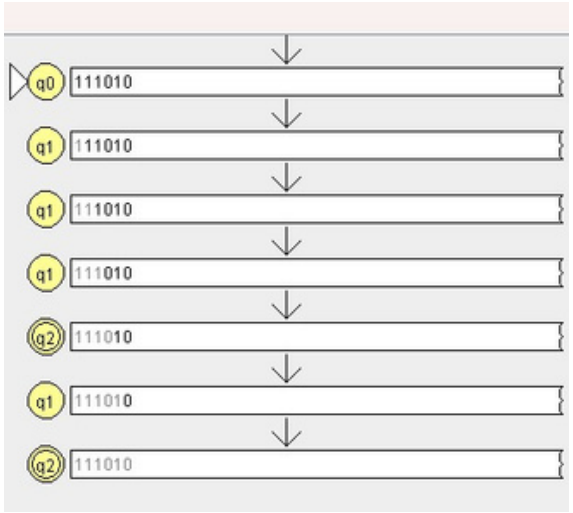


Result: Rejected

2. Design a DFA that accepts all strings ending with 10 over the alphabet Σ = {0,1}

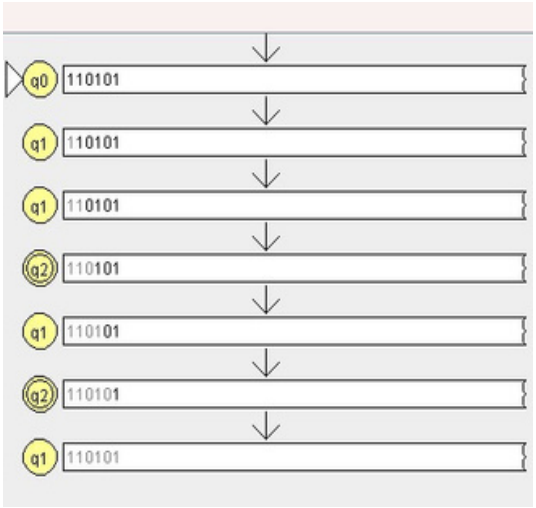


Input: 111010



Result: Accepted

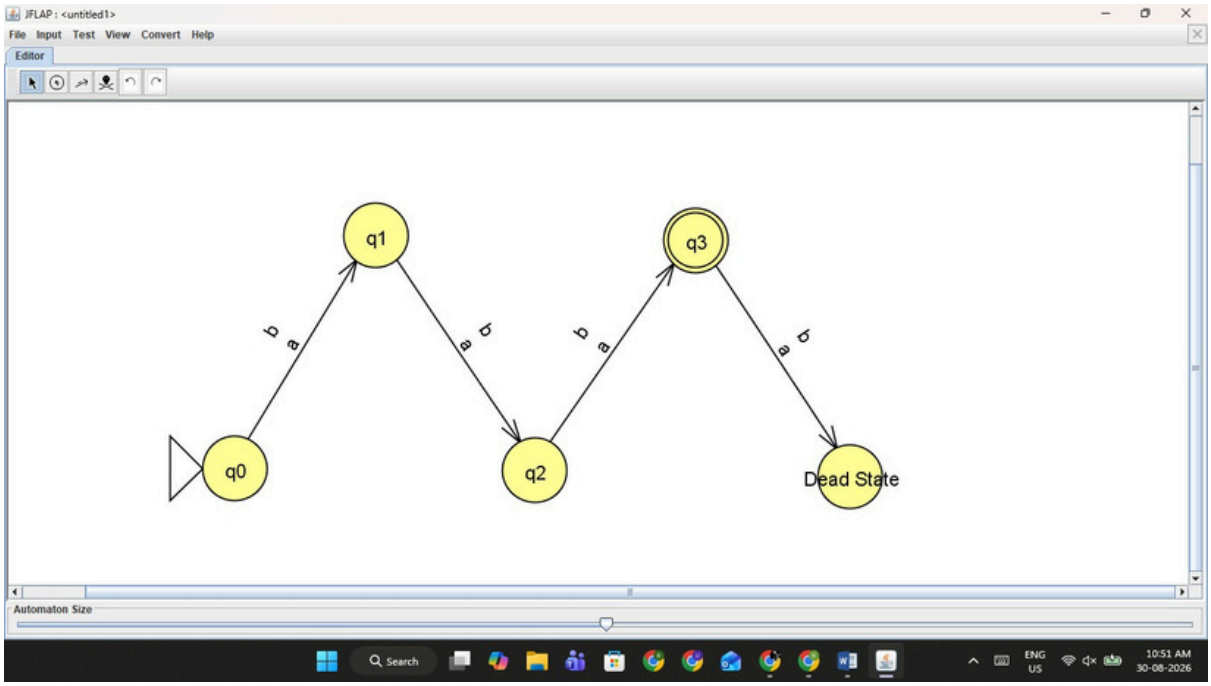
Input: 110101



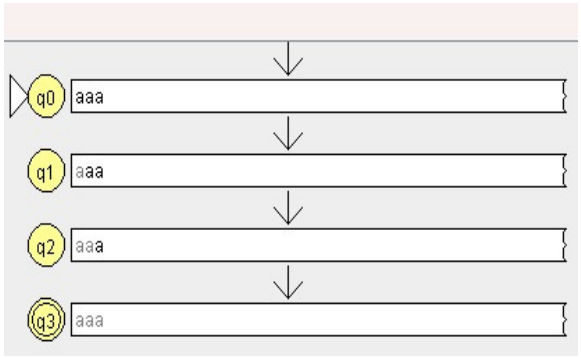
Result: Rejected

3. Design a DFA that accepts all strings having length 3 over the alphabet ∑ = {a,b}

L = { aaa, aba, bbb, aab, abb, bba, bab, baa}

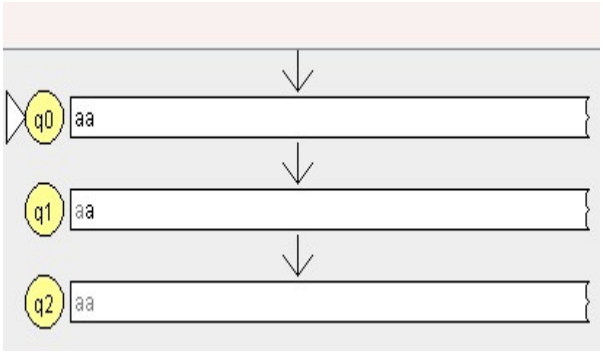


Input: aaa



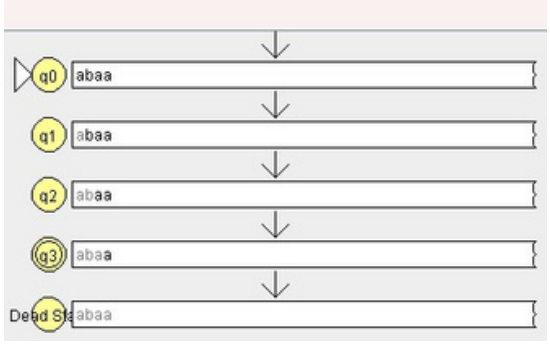
Result: Accepted

Input: aa



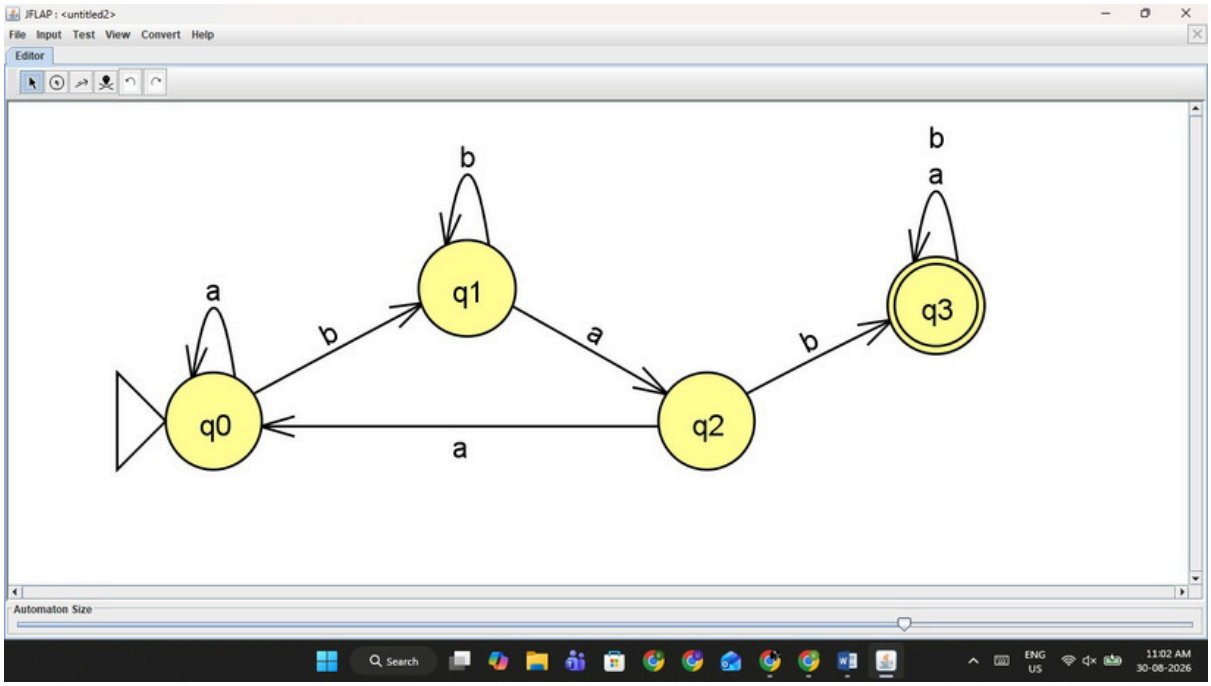
Result: Rejected

Input: abaa

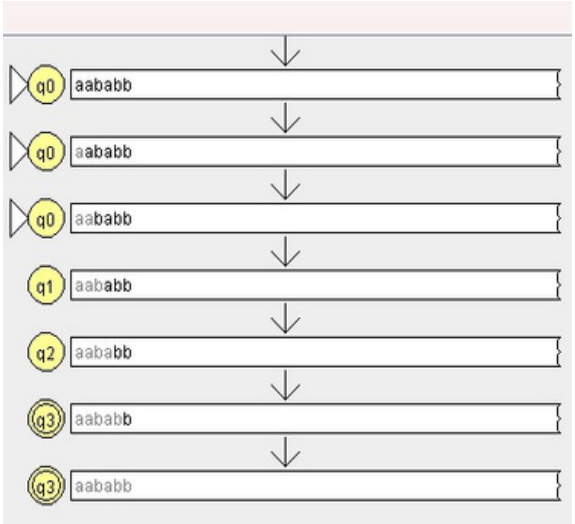


Result: Rejected (Dead state)

4. Design a DFA that accepts all strings having substring ‘bab’ over alphabet Σ = {a,b}

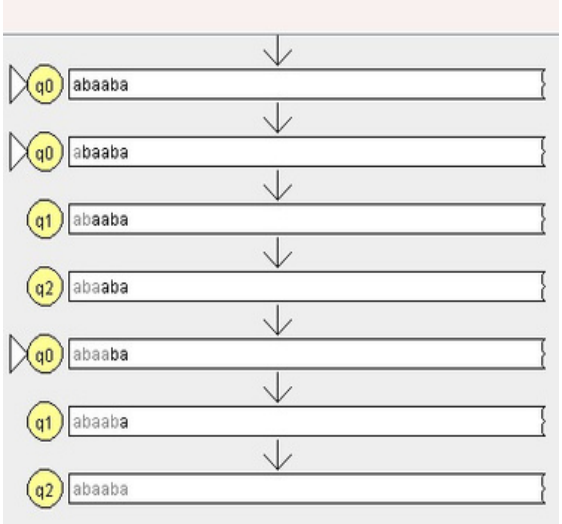


Input: aababb



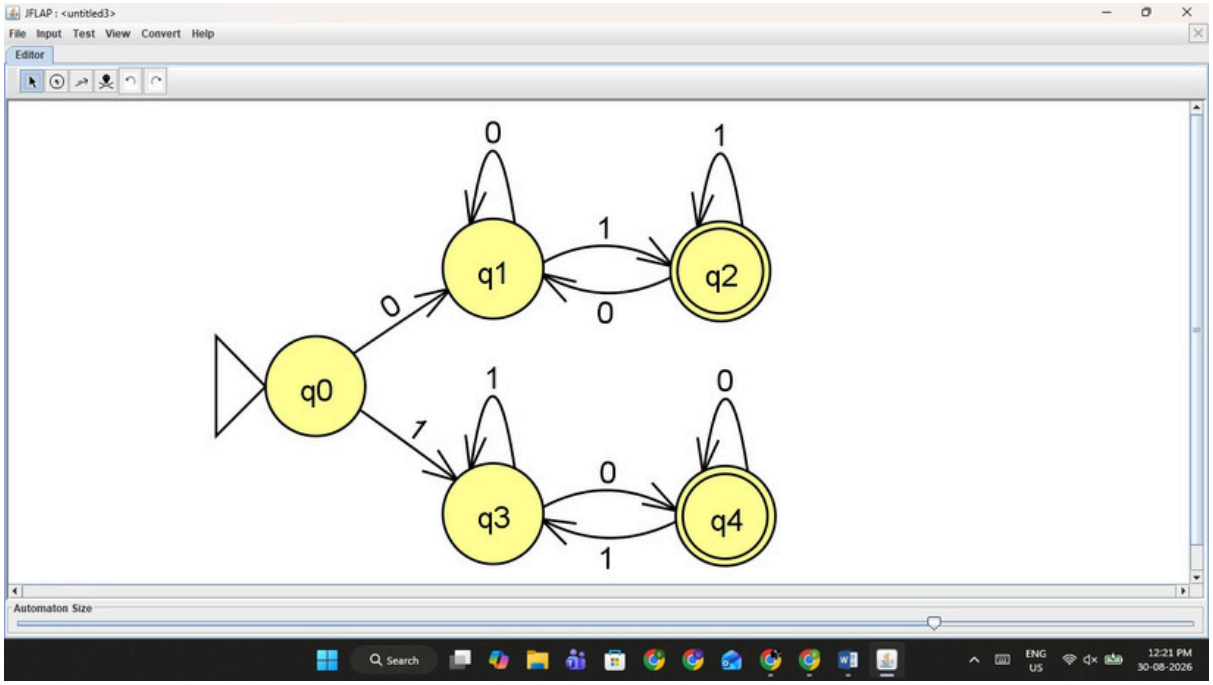
Result: Accepted

Input: abaaba

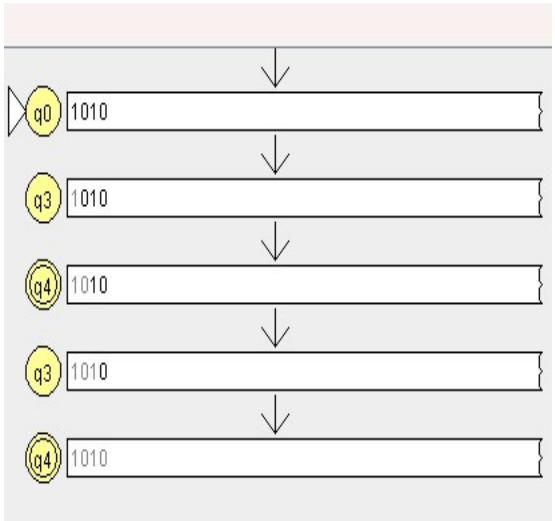


Result: Rejected

5. Draw a DFA for the language accepting strings starting and ending with different characters over the alphabet Σ = {0,1}

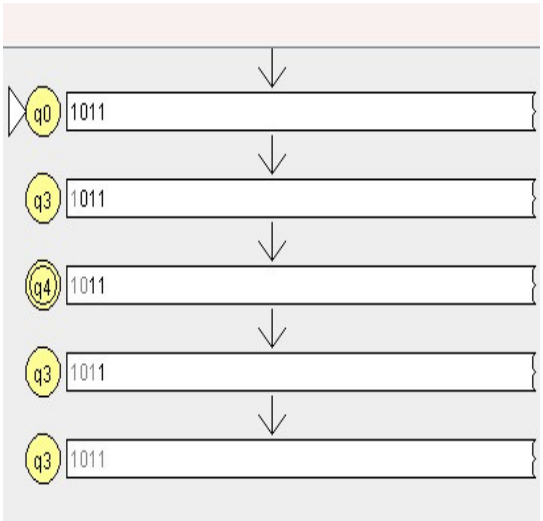


Input: 1010



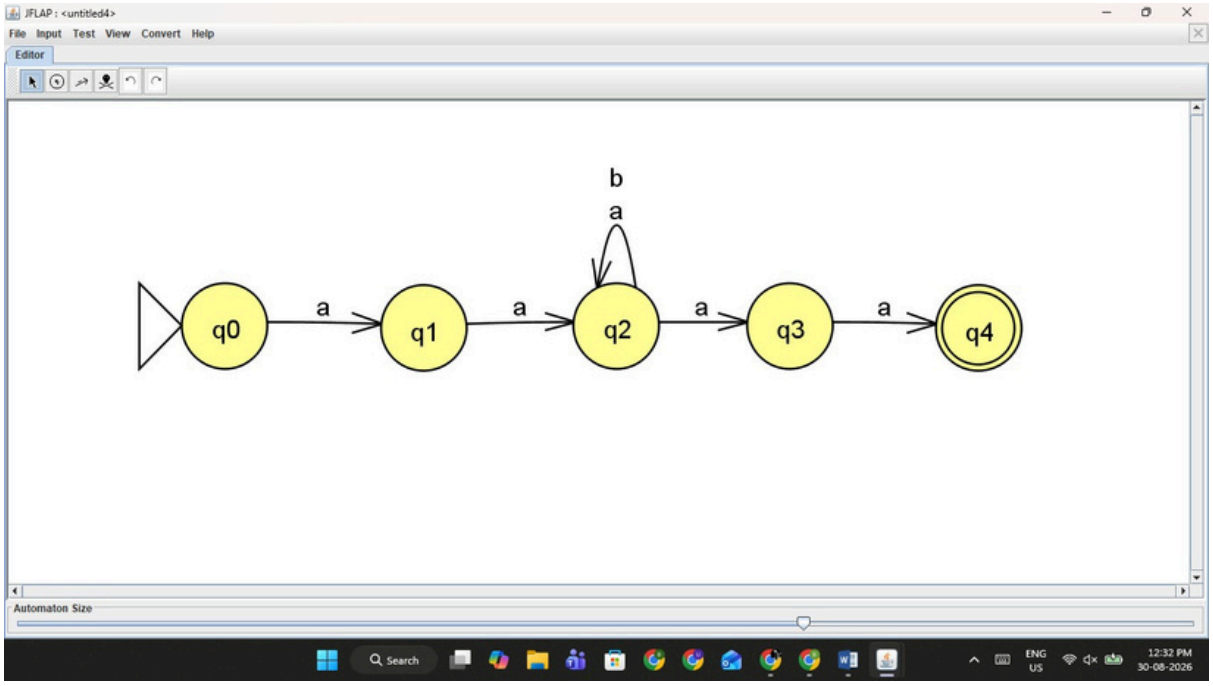
Result: Accepted

Input: 1011

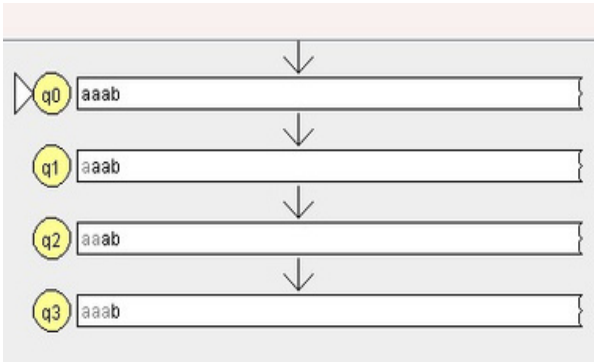


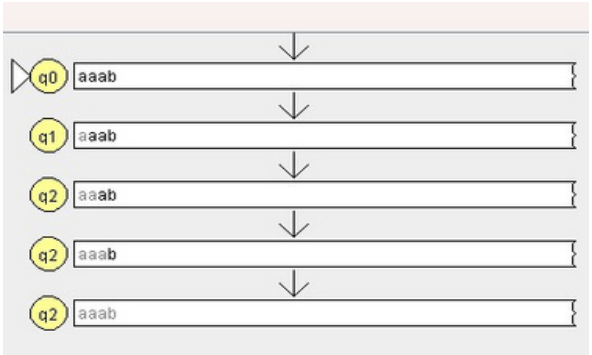
Result: Rejected

6. Design an NFA over the alphabet Σ = {a, b} that accepts all strings that start with `aa` and end with `aa`.



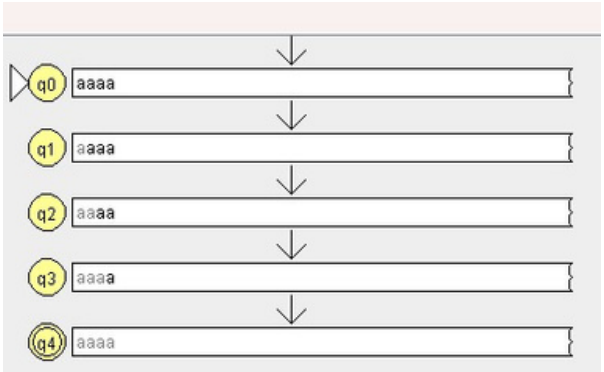
Input: aaab





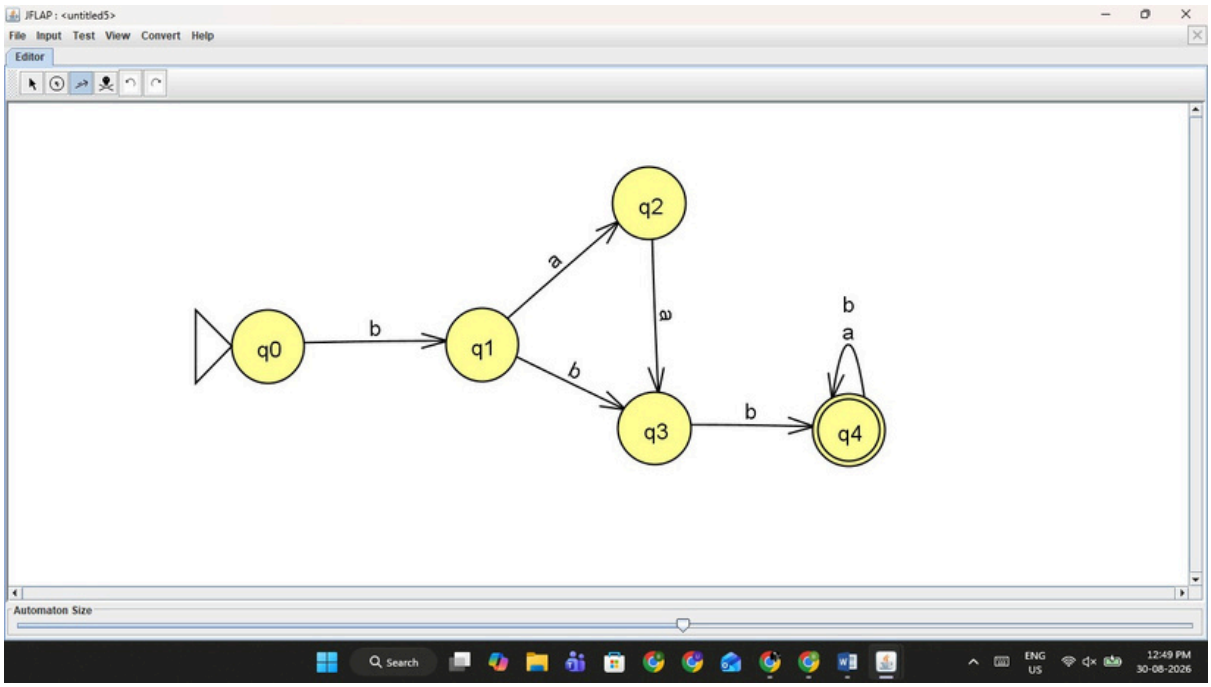
Result: Rejected

Input: aaaa

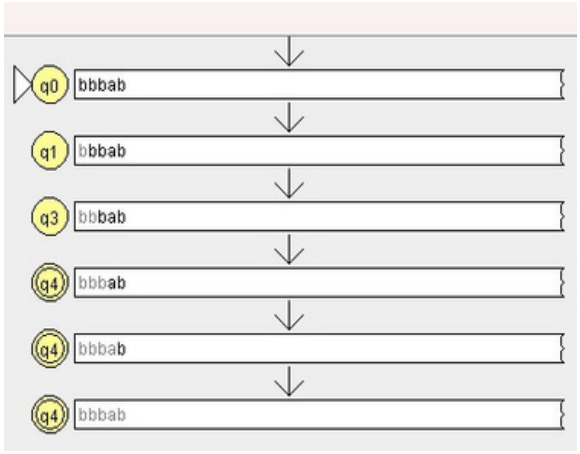


Result: Accepted

7. Design an NFA over the alphabet Σ = {a, b} that accepts all strings that start with `bbb` or `baab`.

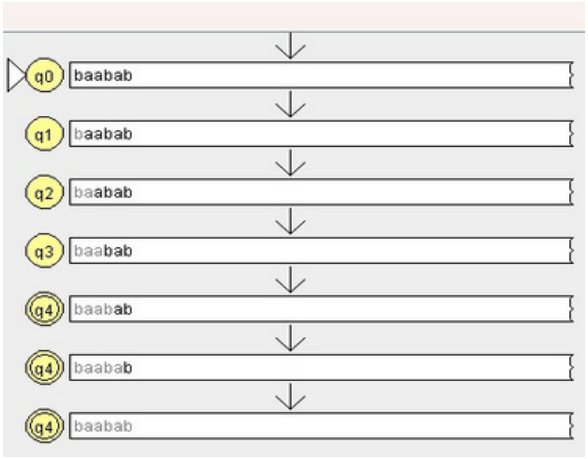


Input: bbbab



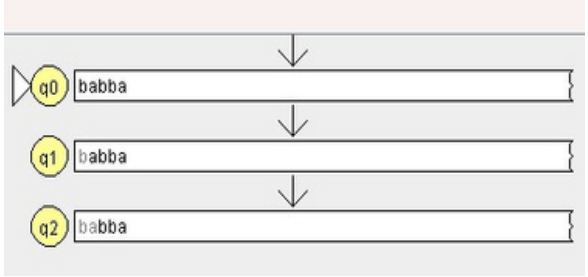
Result: Accepted

Input: baabab



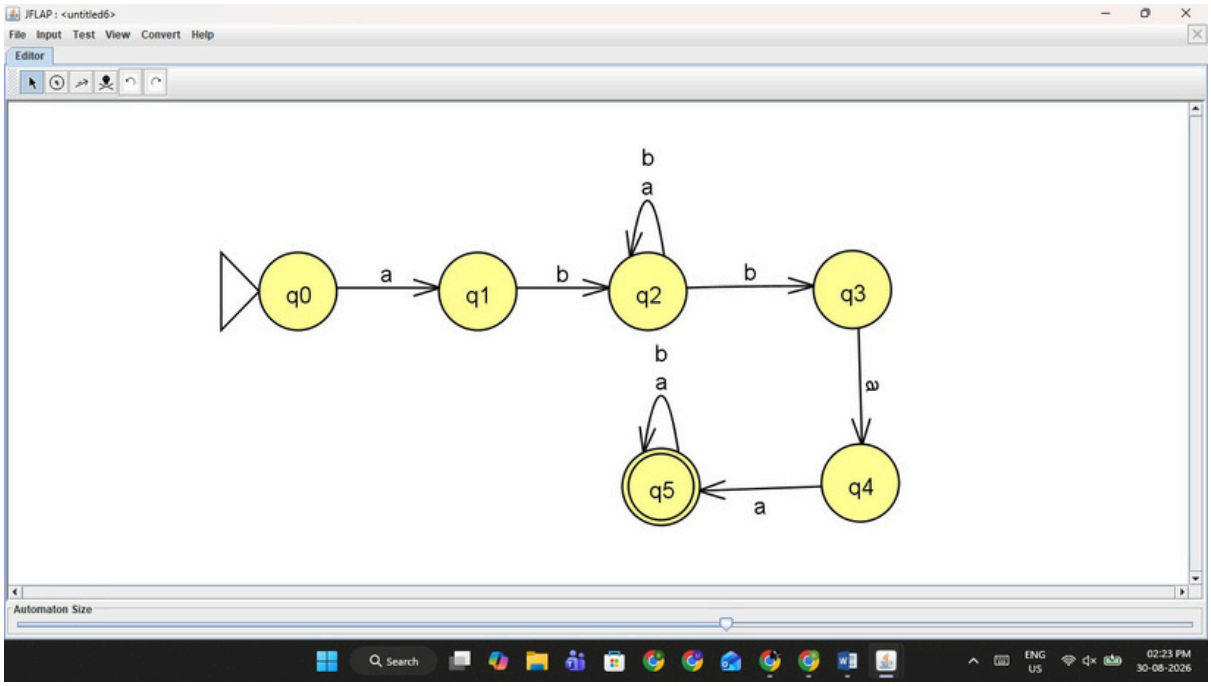
Result: Accepted

Input: babba

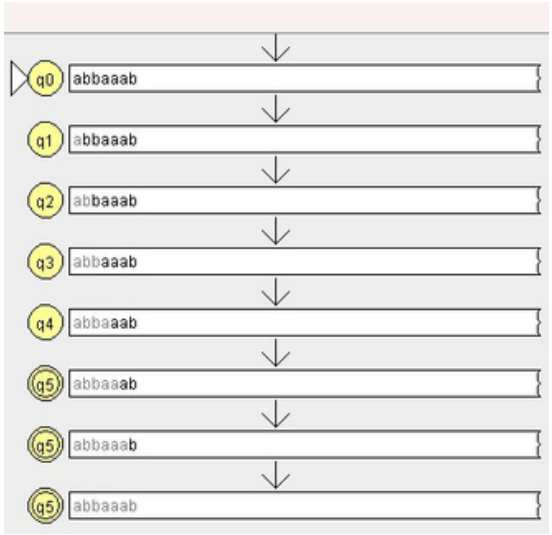


Result: Rejected

8. Construct an NFA over the alphabet Σ = {a, b} that accepts all strings that start with 'ab' and contain 'baa' somewhere after the initial 'ab'.

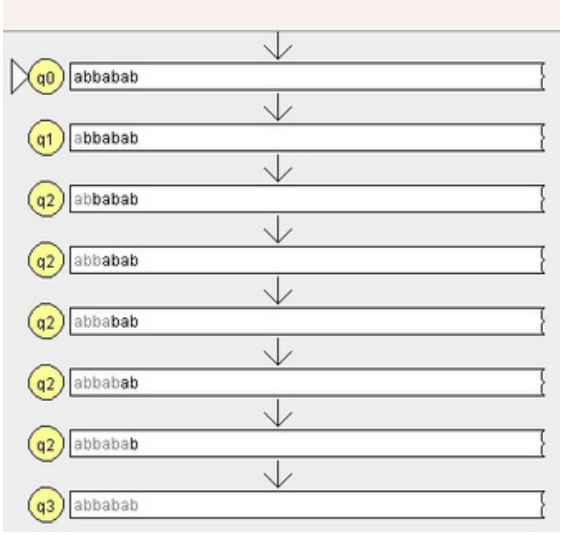


Input: abbaaab



Result: Accepted

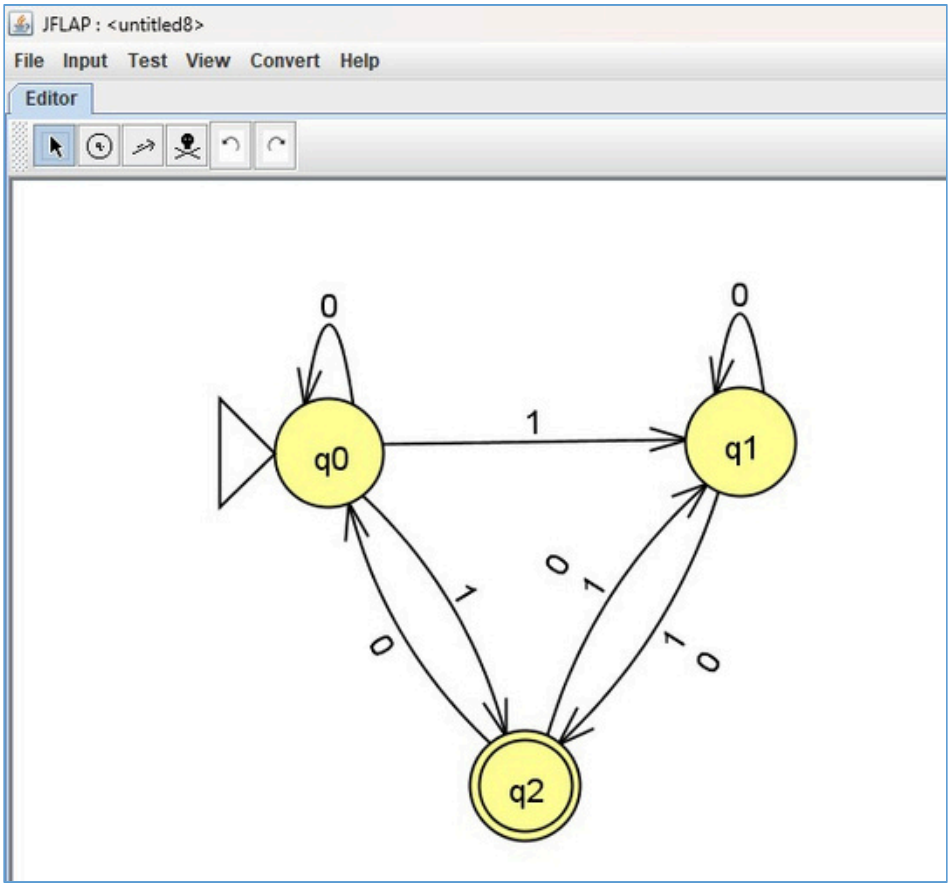
Input: abbabab



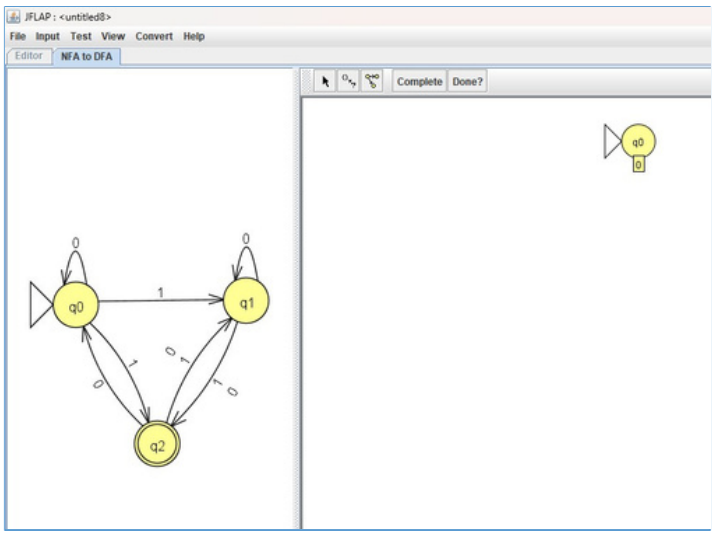
Result: Rejected

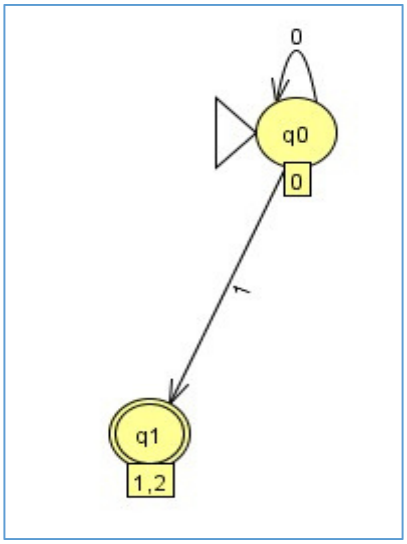
9. Convert NFA to DFA

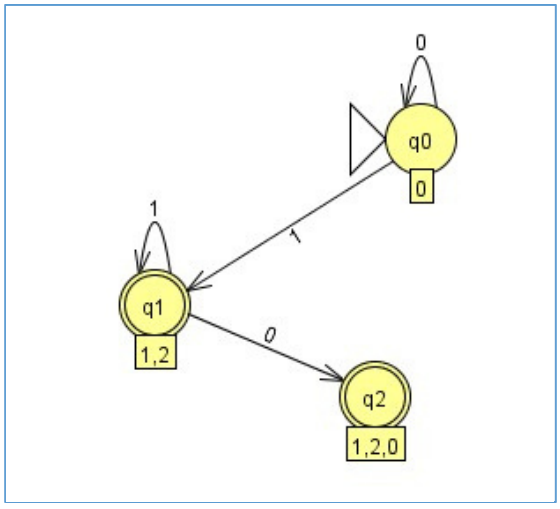
Given NFA:

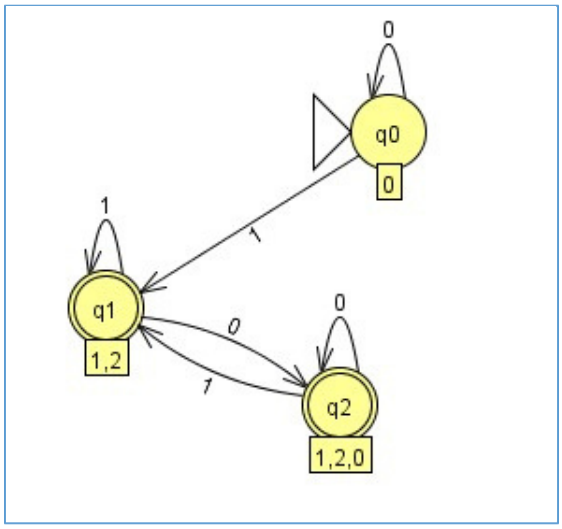


Conversion steps:



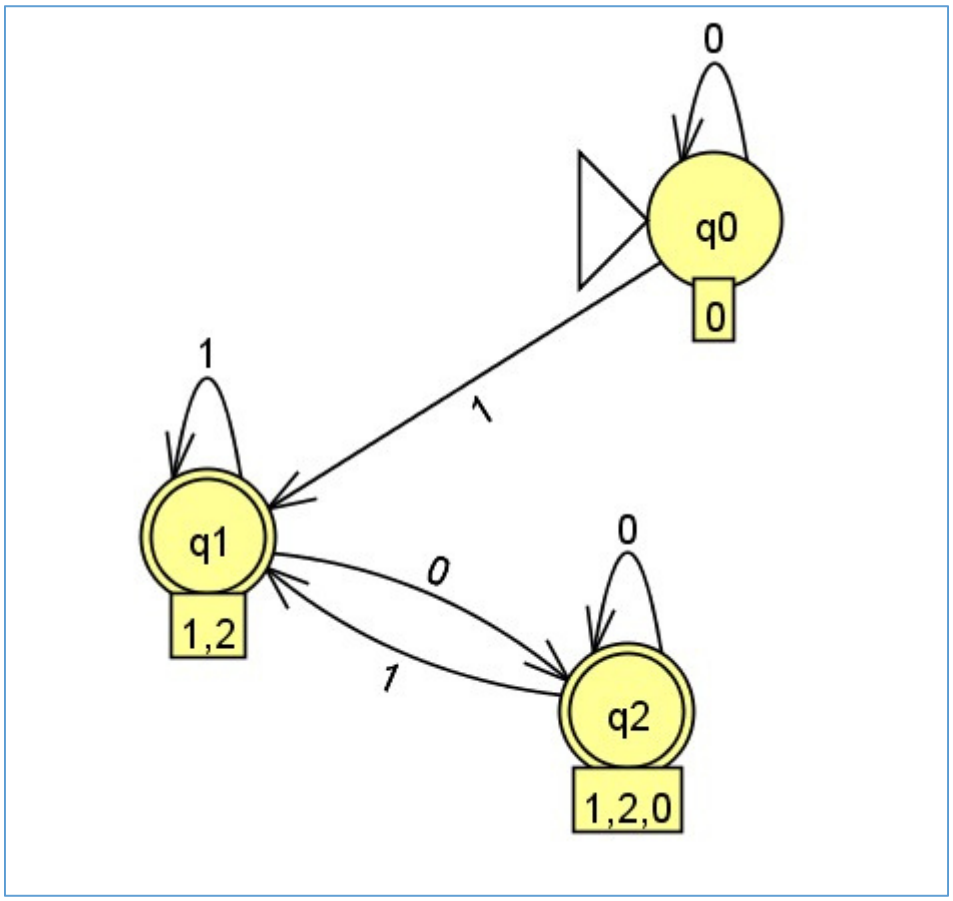






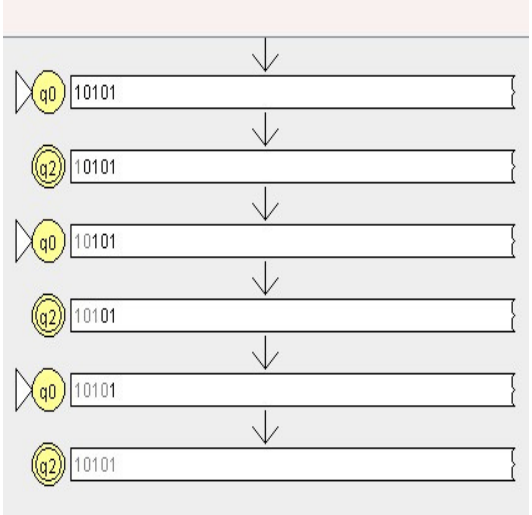


Equivalent DFA:

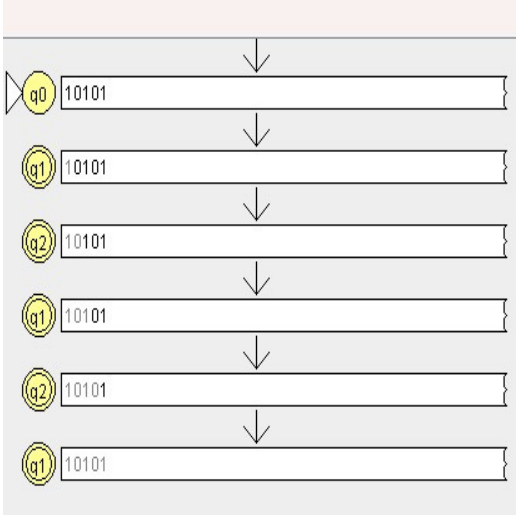


Testing same input on NFA and DFA: 10101

NFA



DFA

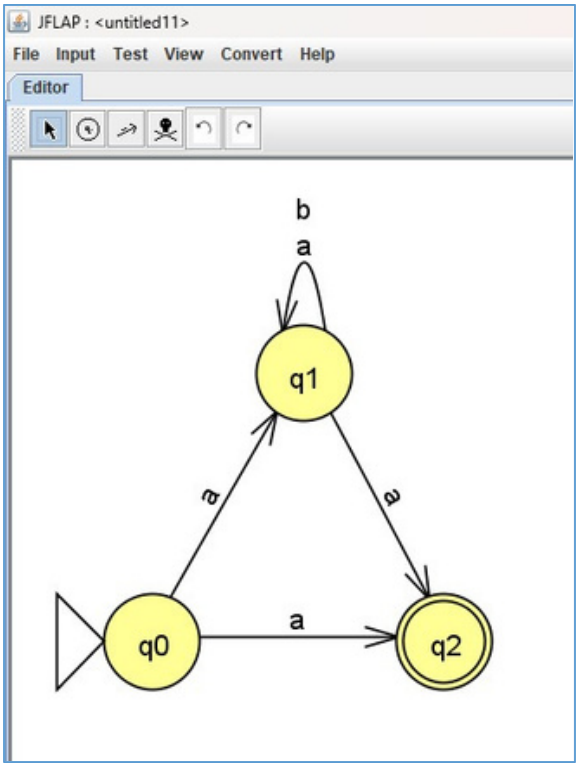


Result: Accepted

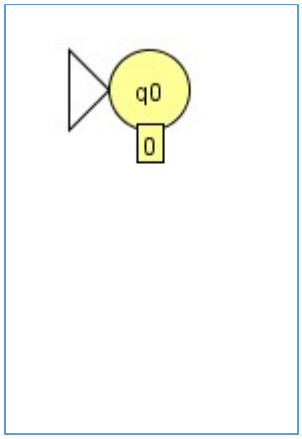
Result: Accepted

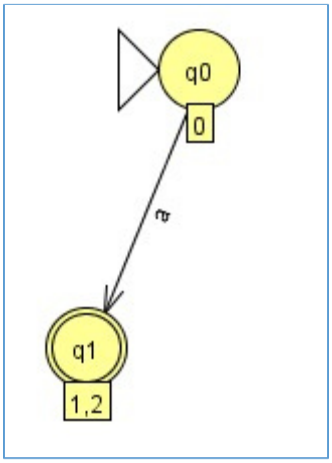
10. Convert NFA to DFA

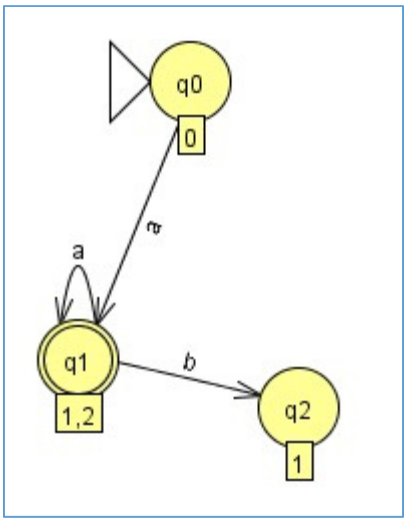
Given NFA

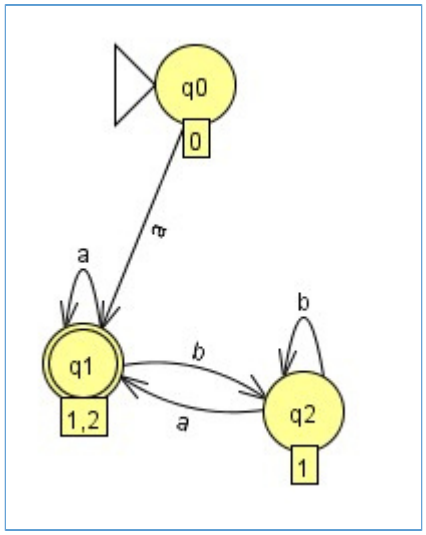


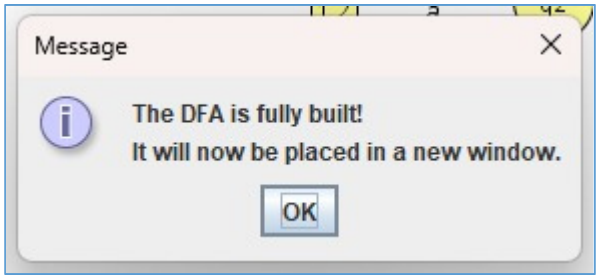
Conversion steps:



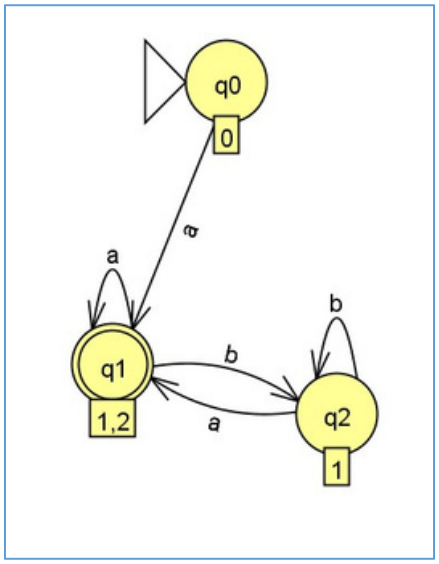






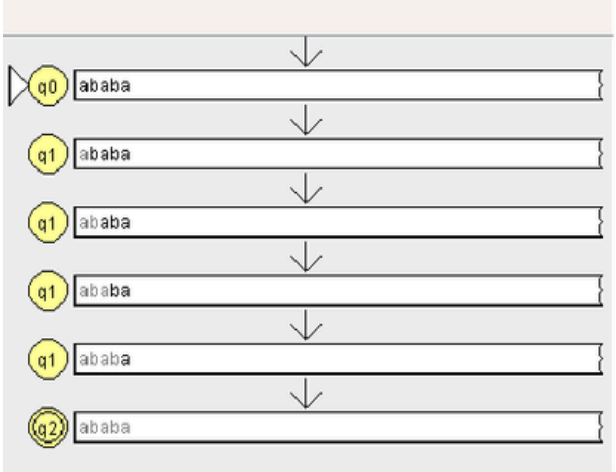


Equivalent DFA:



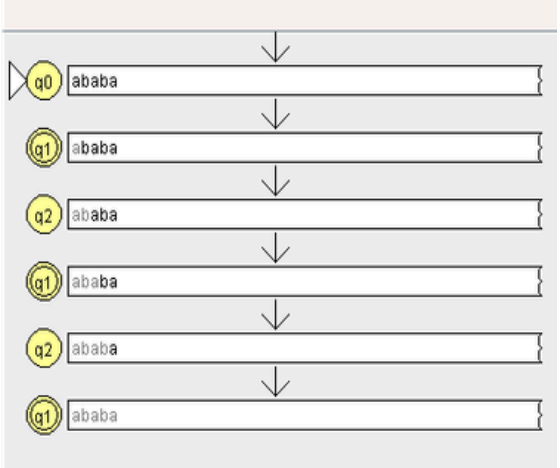
Testing same input on NFA and DFA: ababa

NFA



Result: Accepted

DFA



Result: Accepted